WORK LOG

Monroe Resource Hub | Central Academy of Technology and Arts | TSA 2026

PROJECT:	Monroe Resource Hub	**DATE RANGE:**	January 2025 - January 2026
SCHOOL:	Central Academy of Technology and Arts	**STUDENTS:**	Yatish Grandhe, Dhyan Kanna, Vihaan Kotagiri
ORGANIZATION:	TSA	**STATUS:**	Completed and Deployed

Project Summary

The Monroe Resource Hub is a comprehensive community platform designed to connect residents of Monroe, North Carolina with vital resources, services, and opportunities. The platform includes a resource directory, an AI-powered resume builder, a job application assistant, a community events calendar, and volunteer opportunity listings. Built with Next.js, React, Supabase, and Google Gemini AI, the application is fully deployed at monroeresourcehub.us.

Work Log Entries

Project Planning and Research — January 2025

Hours: 15 hours **Team:** All team members

- Community needs assessment
- Researched existing platforms
- Identified features
- Created timeline and milestones
- Selected tech stack (Next.js, Supabase, Tailwind CSS, TypeScript)

Outcome: Established project scope and technical requirements

UI/UX Design and Wireframing — February 2025

Hours: 20 hours **Team:** Design team

- Wireframes for all pages
- Color scheme and branding
- Custom logo design
- Responsive layouts
- Design system and component library

Outcome: Complete design system and visual mockups

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Database Setup and Backend Development	March - April 2025

Hours: 25 hours **Team:** Backend team

- Database schema for resources/events/users
- Supabase project setup
- Authentication system
- API routes
- Row Level Security (RLS) policies

Outcome: Fully functional backend with secure authentication

Frontend Development - Core Pages	May - June 2025

Hours: 30 hours **Team:** Frontend team

- Homepage with hero and categories
- Resource directory with search and filters
- Resource detail pages
- Events calendar with React Big Calendar
- About page and contact forms
- Responsive navigation and footer

Outcome: Complete user-facing pages and navigation

Career Center Development	July - September 2025

Hours: 35 hours **Team:** AI/Backend team

- Google Gemini AI integration
- AI-powered resume builder with templates
- Job application assistant
- Local job board
- PDF export with jspdf and html2canvas

Outcome: Fully functional career center with AI-powered features

Testing and Quality Assurance	November 2025

Hours: 18 hours **Team:** All team members

- Cross-browser testing (Chrome, Firefox, Edge)
- Responsive device testing
- Security and vulnerability scans
- Full user flow testing
- Bug fixes and performance improvements
- Accessibility validation

Outcome: Stable, tested, and secure application

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Content Population and Data Entry	December 2025

Hours: 15 hours **Team:** Content team

- 50+ community resources added
- Organized by category
- Event listings and calendar entries
- Job listings
- Wrote all page content

Outcome: Fully populated database with real community data

Documentation and Final Preparation	January 2026

Hours: 12 hours **Team:** All team members

- Reference page with all sources
- Student copyright checklist
- Work log entries
- Presentation materials
- User documentation
- Final review and quality check

Outcome: Complete project documentation and competition-ready submission

Total Time Summary

Phase	Date Range	Hours	Team Members
1. Project Planning and Research	January 2025	15	All team members
2. UI/UX Design and Wireframing	February 2025	20	Design team
3. Database Setup and Backend	March - April 2025	25	Backend team
4. Frontend - Core Pages	May - June 2025	30	Frontend team
5. Career Center Development	July - September 2025	35	AI/Backend team
6. Testing and QA	November 2025	18	All team members
7. Content Population	December 2025	15	Content team
8. Documentation and Final Prep	January 2026	12	All team members
TOTAL	**Jan 2025 - Jan 2026**	**170**	**25+ students**

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Team Contributions

Collaborative project by the CATA TSA Chapter. Student developers led development, design, and content.

Name	Role
Yatish Grandhe	Student Developer
Dhyan Kanna	Student Developer
Vihaan Kotagiri	Student Developer

PROJECT COMPLETION:	January 2026	**STATUS:**	Completed and Deployed
DEPLOYMENT:	Vercel Platform	**URL:**	https://monroeresourcehub.us